Scalable fit of the output-error sensitivity of a Clements unitary mesh vs. mean insertion loss ($\sigma_L = 0$)

FIT on the small range $[0, 1]$ dB

Model (overlap-compatible, scalable in N/N_0): $E_L(\mu_L, N) = \sqrt{2}\,\sqrt{1 - e^{-u_L^2}}$, $u_L = \dfrac{s_L(N)\,\mu_L + \gamma_L(N)\,\mu_L^2}{\sqrt{2}}$, $s_L(N) = s_{L,0}\,(N/N_0)^{p_L}$, $\gamma_L(N) = \gamma_0\,(N/N_0)^{q_L}$, $N_0 = 100$

$s_{L,0}$ = initial linear sensitivity to loss at $N_0 = 100$ (dB^{-1}) | p_L = network-size exponent of s_L

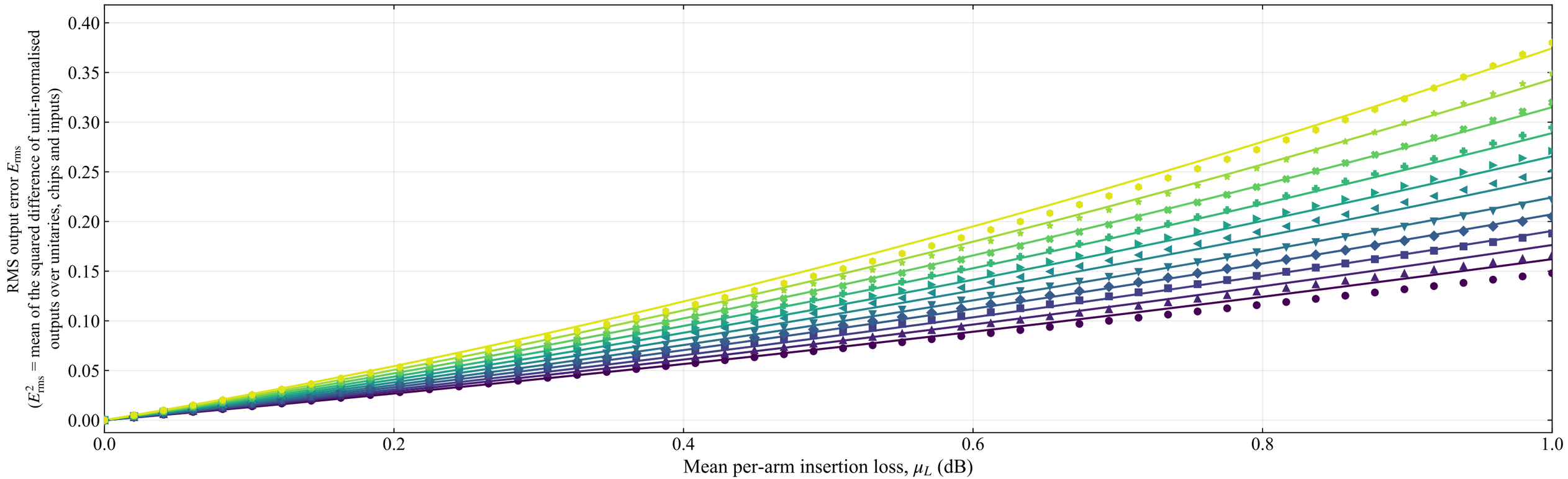
γ_0 = quadratic (initial-convexity) coefficient at $N_0 = 100$ (dB^{-2}) | q_L = network-size exponent of γ_L

Best-fit values (small range, all N, equal weight per point; $\pm$ = bootstrap s.d. over unitaries): $s_{L,0} = 0.17639 \pm 0.00089\ \mathrm{dB}^{-1}$, $p_L = 0.20334 \pm 0.00670$, $\gamma_0 = 0.06960 \pm 0.00095\ \mathrm{dB}^{-2}$, $q_L = 0.41848 \pm 0.01497$

Fit quality (fit range): all 550 points: RMSE = 0.00345, MAE = 0.00259, $R^2 = 0.99826$, $\max|\Delta E| = 0.0141$ | off-plateau ($E \le 1.2$, 550 points): RMSE = 0.00345, $R^2 = 0.99826$, mean rel. err. = 3.12%

Hardware model: per-arm loss $L \sim \mathrm{TruncNorm}(\mu_L, \sigma_L^2; 0, 2\mu_L)$ dB with $\sigma_L = 0$ (deterministic loss); all other channels ideal | markers = simulated E_{rms} (over unitaries, chips and input types), solid lines = scalable fit (fit in E space)

$N = \{20, 28, 38, 53, 72, 100, 138, 190, 263, 362, 500\}$ | Haar-random unitary U (single Clements mesh) | 7 input types × 10 draws, pooled (equal weight) | one forward pass $y = Ux$ | $E_{\mathrm{rms}} = \sqrt{\langle \|\hat{y}^{\mathrm{hw}} - \hat{y}^{\mathrm{ideal}}\|_2^2 \rangle}$ | 25 unitaries × 30 chips | 50 points per curve



Network size N | scaled sensitivity from the fit (markers: data, lines: fit)

- $N = 20$, $s_L = 0.127\ \mathrm{dB}^{-1}$, $\gamma_L = 0.035\ \mathrm{dB}^{-2}$
- $N = 28$, $s_L = 0.136\ \mathrm{dB}^{-1}$, $\gamma_L = 0.041\ \mathrm{dB}^{-2}$
- $N = 38$, $s_L = 0.145\ \mathrm{dB}^{-1}$, $\gamma_L = 0.046\ \mathrm{dB}^{-2}$
- $N = 53$, $s_L = 0.155\ \mathrm{dB}^{-1}$, $\gamma_L = 0.053\ \mathrm{dB}^{-2}$
- $N = 72$, $s_L = 0.165\ \mathrm{dB}^{-1}$, $\gamma_L = 0.061\ \mathrm{dB}^{-2}$
- $N = 100$, $s_L = 0.176\ \mathrm{dB}^{-1}$, $\gamma_L = 0.070\ \mathrm{dB}^{-2}$
- $N = 138$, $s_L = 0.188\ \mathrm{dB}^{-1}$, $\gamma_L = 0.080\ \mathrm{dB}^{-2}$
- $N = 190$, $s_L = 0.201\ \mathrm{dB}^{-1}$, $\gamma_L = 0.091\ \mathrm{dB}^{-2}$
- $N = 263$, $s_L = 0.215\ \mathrm{dB}^{-1}$, $\gamma_L = 0.104\ \mathrm{dB}^{-2}$
- $N = 362$, $s_L = 0.229\ \mathrm{dB}^{-1}$, $\gamma_L = 0.119\ \mathrm{dB}^{-2}$
- $N = 500$, $s_L = 0.245\ \mathrm{dB}^{-1}$, $\gamma_L = 0.136\ \mathrm{dB}^{-2}$